

Import Data using Transform Maps (Spreadsheet)

Project Description:

The project is developed on the ServiceNow platform to simplify and improve employee data management by providing an efficient way to import, store, manage, and monitor employee information. In many situations, entering employee details manually into a system can take more time and may lead to incorrect or missing information. To overcome this problem, the project uses a spreadsheet as the source of employee data and imports the information into ServiceNow using **Import Sets and Transform Maps**. The spreadsheet contains employee details such as **Employee ID, Employee Name, Email, Department, and Location**. The Import Set is used to bring the spreadsheet data into ServiceNow, while the Transform Map is used to map the imported fields to the corresponding fields in the custom **Employee Test** table.

After the transformation process, employee records are created and stored in the Employee Test table, where the data can be viewed and managed easily. The project also provides reporting features to help users understand and monitor the employee information. Reports such as **Employees by Department, Employee by Location, and Employee List Report** are created to present employee data in different formats. These reports make it easier to view employee information, identify department and location details, and check the available employee records. In addition, an **Employee Analytics Dashboard** is created to bring the reports together in a single place, allowing users to view employee information and analytics more conveniently. The dashboard provides a simple visual representation of the employee data and helps users monitor the information without checking each report separately. The project demonstrates the practical use of important ServiceNow features such as **custom tables, Import Sets, Transform Maps, Reports, and Dashboards** for employee data management. It also shows how spreadsheet-based employee information can be transformed into structured records within ServiceNow and then used for reporting and analytics. Overall, the project aims to reduce manual data entry, save time, improve the accuracy and consistency of employee records, and make employee information easier to access, manage, analyze, and monitor. By combining data import, record management, reporting, and dashboard functionality in one ServiceNow-based solution, the project provides a structured approach for handling employee information efficiently.

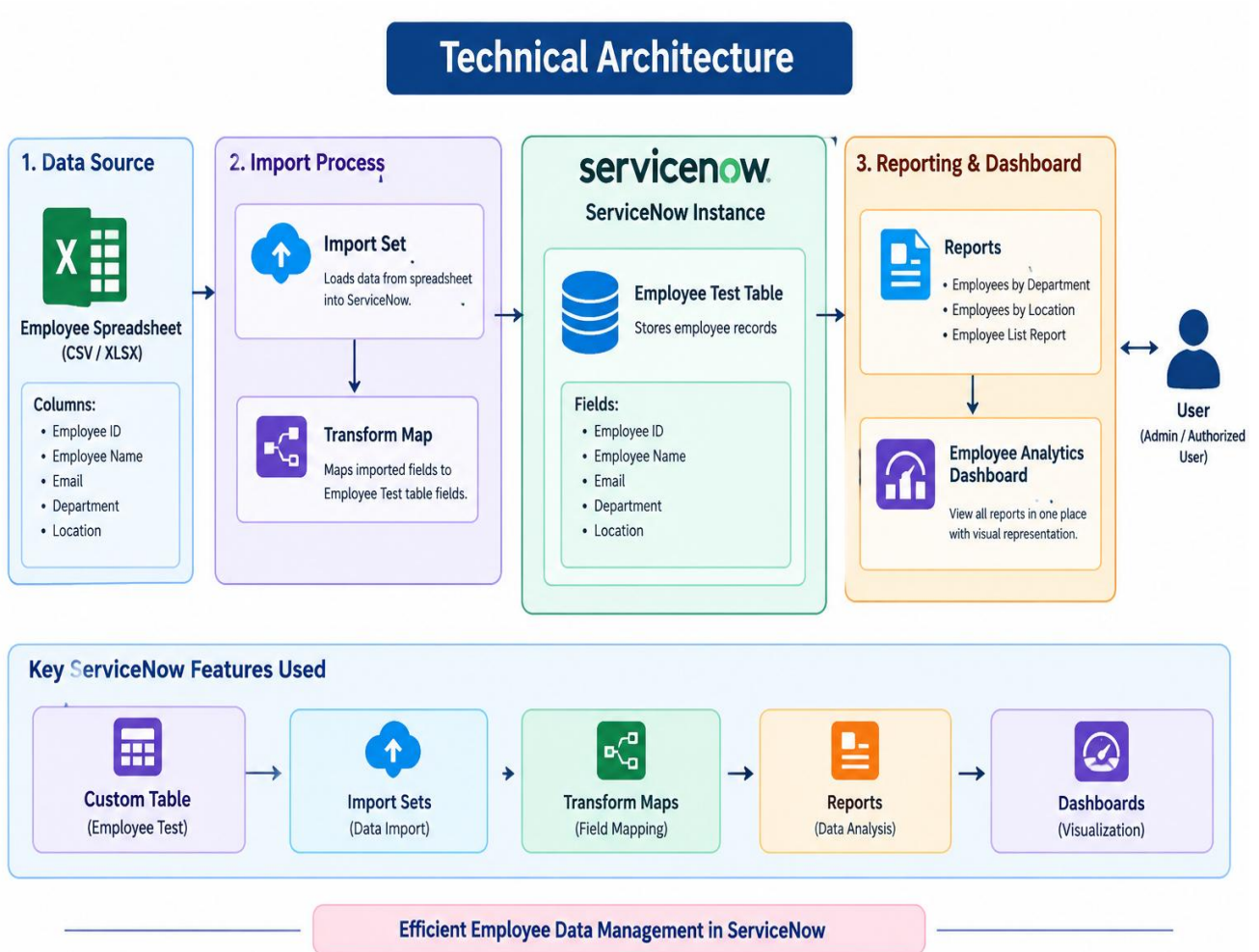
Scenarios

Scenario 1: A user describes a new summer collection of sustainable sunglasses and selects Instagram with a Casual tone. The system generates three emoji-rich captions optimized for Instagram's visual-first audience, ten relevant hashtags mixing broad and niche tags, and three punchy CTAs all in seconds.

Scenario 2: A user promoting a B2B SaaS product selects LinkedIn with a Professional tone. CaptionAI generates thought-leadership-style captions with a value-driven narrative, industry-relevant hashtags, and CTAs designed to drive professional engagement such as demo sign-ups or article reads.

Scenario 3: A user running a weekend flash sale selects Instagram with a Promotional tone. The AI generates urgency-driven captions filled with power words, a curated hashtag set targeting deal-seekers, and CTAs like “Shop before it’s gone!” to maximize conversion.

Technical Architecture:



Description: CaptionAI utilizes a modular three-tier architecture to deliver rapid, AI-driven content generation. The frontend provides a responsive user interface built with HTML and CSS while the Flask-based backend orchestrates data validation and prompt construction. It interfaces with the Groq API using the LLaMA 3.3-70B model to generate tailored social media content, with secure deployment managed locally via Flask and exposed publicly through Ngrok.

(Note: If you are using database in your project definitely you need to add ER Diagram along with description)

Pre-requisites:

- **Flask Framework Knowledge:** [Flask Documentation](#)
- **Groq API Familiarity:** [Groq API Documentation](#)
- **HTML, CSS, and JavaScript Skills:** [W3Schools HTML/CSS/JavaScript Tutorials](#)
- **Python Programming Proficiency:** [Python Documentation](#)
- **Version Control with Git:** [Git Documentation](#)
- **Development Environment Setup:** [Flask Installation Guide](#)
- **Ngrok for Public Deployment:** [Ngrok Documentation](#)

Project Workflow:

Milestone 1: Model Selection and Architecture

- **Activity 1.1:** Generate a Groq API key and configure it securely in the project environment.
- **Activity 1.2:** Research and select the appropriate generative AI model from Groq for social media content generation.
- **Activity 1.3:** Define the architecture of the application, detailing interactions between the frontend, backend, and AI integration.
- **Activity 1.4:** Set up the development environment, installing necessary libraries and dependencies for Flask and the Groq API.

Milestone 2: Core Functionalities Development

- **Activity 2.1:** Develop the core functionalities: Caption Generator, Hashtag Suggester, and Call-to-Action Generator.

- **Activity 2.2:** Implement the Flask backend to manage routing and user input processing, ensuring smooth API interactions.

Milestone 3: App.py Development

- **Activity 3.1:** Write the main application logic in app.py, establishing routes for each feature and integrating AI responses.

Milestone 4: Frontend Development

- **Activity 4.1:** Design and develop the user interface using HTML, CSS, and JavaScript, ensuring a responsive and intuitive layout.
- **Activity 4.2:** Create dynamic templates with Flask's render_template to display results based on user interactions.

Milestone 5: Deployment

- **Activity 5.1:** Prepare the application for local deployment by configuring the server environment and ensuring all dependencies are correctly installed.
- **Activity 5.2:** Deploy the application publicly using Ngrok to expose the local Flask server over a secure public URL.

Milestone 6: Conclusion

Milestone 1: Creation Of Spread sheet and Tabel

In this milestone, we focus on selecting the appropriate generative AI model from Groq for our social media content generation needs. This involves researching the capabilities and performance of various models that Groq offers, ensuring that the chosen model aligns well with the application's objectives of generating captions, hashtags, and CTAs across different platforms and tones.

Creation Of Spreadsheet

- | | |
|---|---|
| • | Open New Google Spreadsheet |
| • | Create New Google Spreadsheet with sample data as shown below |

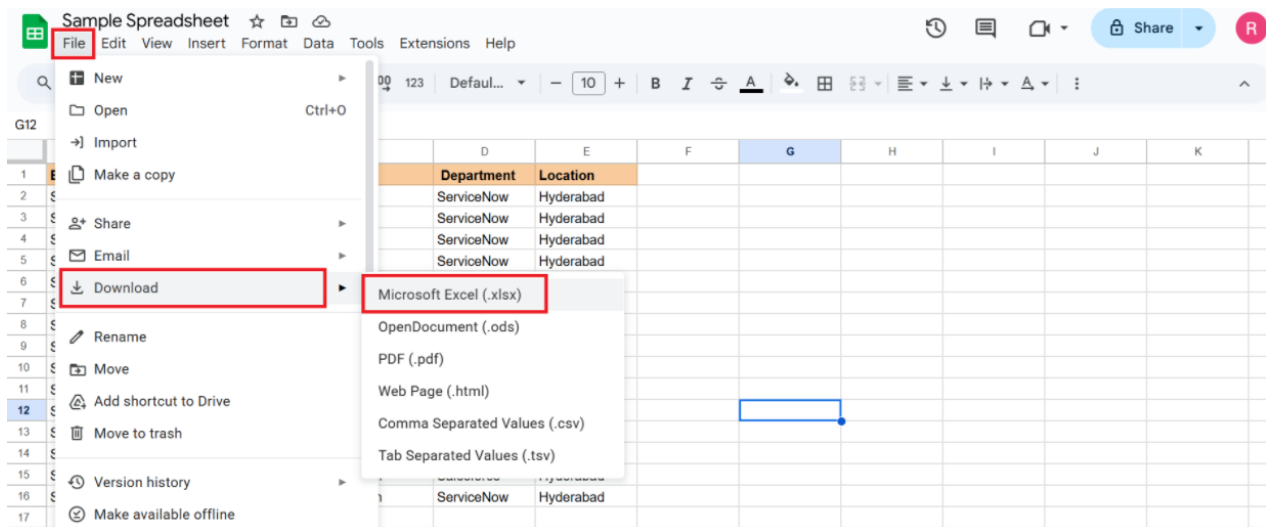
Sample Spreadsheet

File Edit View Insert Format Data Tools Extensions Help

100% 123 Default... 10 B I A

Employee ID	Name	Email	Department	Location
SB-0001	Ratan Kumar	test1@gmail.com	ServiceNow	Hyderabad
SB-0002	Sandeep Guja	test2@gmail.com	ServiceNow	Hyderabad
SB-0003	Meghana Katoju	test3@gmail.com	ServiceNow	Hyderabad
SB-0004	Ajay Kumar	test4@gmail.com	ServiceNow	Hyderabad
SB-0005	Bitla Sai Prasad	test5@gmail.com	ServiceNow	Hyderabad
SB-0006	Sushma Ganji	test6@gmail.com	ServiceNow	Hyderabad
SB-0007	Yaswanth Nikku	test7@gmail.com	ServiceNow	Hyderabad
SB-0008	Prashanth Reddy	test8@gmail.com	ServiceNow	Hyderabad
SB-0009	Praneeth S	test9@gmail.com	ServiceNow	Hyderabad
SB-0010	Ravi Teja	test10@gmail.com	ServiceNow	Hyderabad
SB-0011	Abhishek S	test11@gmail.com	Salesforce	Hyderabad
SB-0012	Rakesh Bhoomani	test12@gmail.com	Salesforce	Hyderabad
SB-0013	Mounika K	test13@gmail.com	Salesforce	Hyderabad
SB-0014	Sushanth J	test14@gmail.com	Salesforce	Hyderabad
SB-0015	Srinu Nandipalli	test15@gmail.com	ServiceNow	Hyderabad

Download the spreadsheet as Sample Spreadsheet.xlsx format.



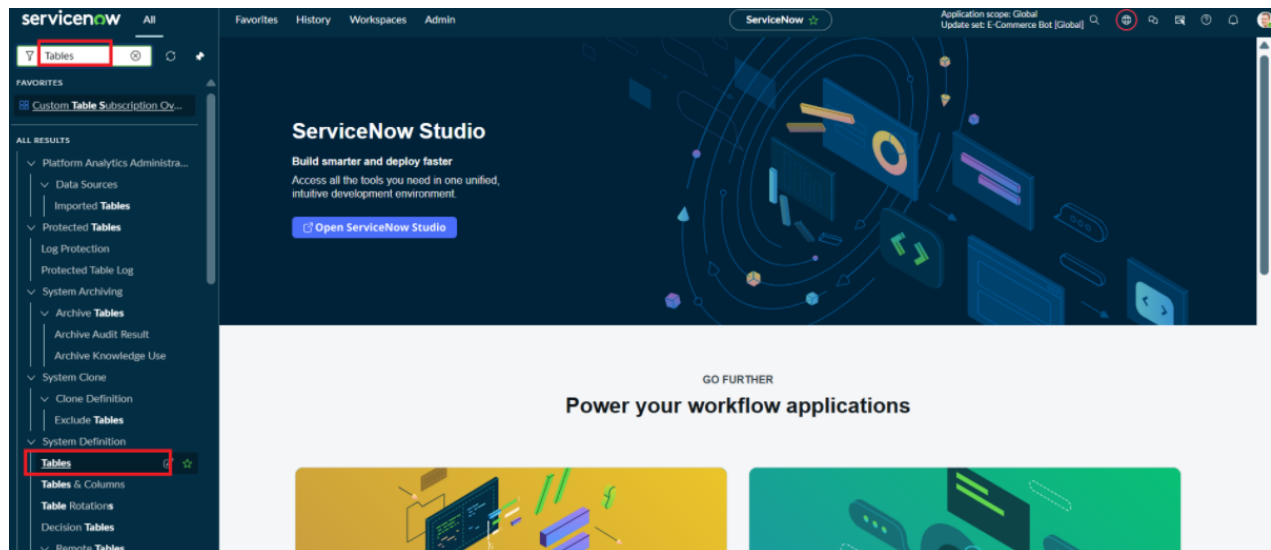
Creation of Tables

The **Import Set Table** acts as a temporary staging table for the data uploaded from the spreadsheet. The imported spreadsheet records are stored here before transformation.

Create new Custom table

Navigation:

Tables > Create New



Label: Employee Test Name: u_employee_test

After creating and saving the Employee Test table

scroll down and go to related links

click on show form

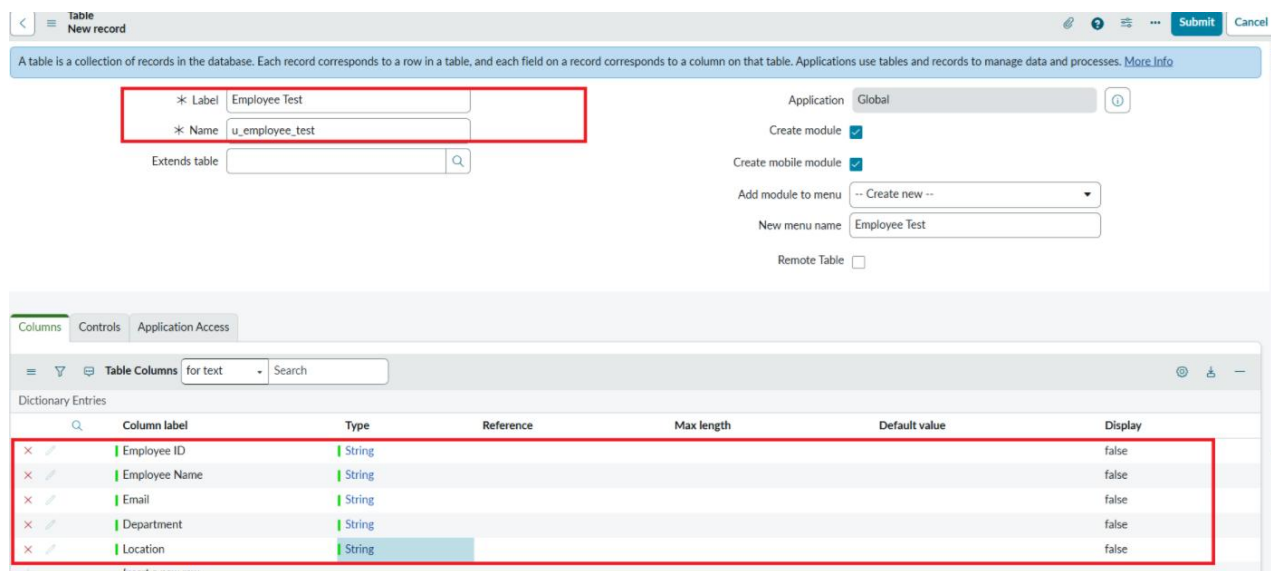


Table: Employee Test

Name: u_employee_test

Application: Global

Create module: ☒

Create mobile module: ☒

Add module to menu: -- Create new --

New menu name: Employee Test

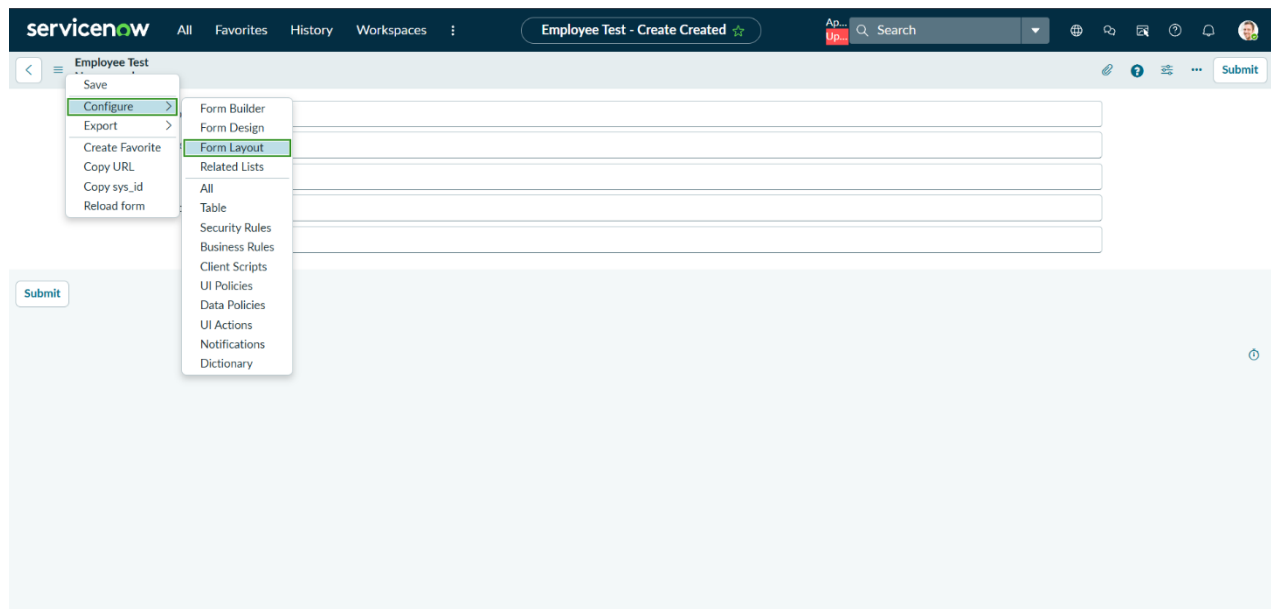
Remote Table: ☐

Column label	Type	Reference	Max length	Default value	Display
Employee ID	String				false
Employee Name	String				false
Email	String				false
Department	String				false
Location	String				false

On the form, locate the **Form Context Menu / Additional Actions** option on the left side of the form header.

Click the **Form Context Menu / Additional Actions** icon.

Click configure , select **Form Layout**.



The **Form Layout** page will open, where you can configure the fields displayed on the form

After opening **Form Layout**, click **New** to create a new field.

Enter the field details:

Column Label: Employee ID

Type: String

Click **Add** to create the field.

Repeat the same steps to create the following fields:

Employee Name – Type: String

Email – Type: String

Department – Type: String

Location – Type: String

After creating all the required fields, click **Save**.

Verify that the newly created fields are displayed on the **Employee Test** form

[<](#) [Employee Test](#)
New record [🔗](#) [🔍](#) [⚙️](#) [⋮](#) [Submit](#)

Employee ID

Employee Name

Email

Department

Location

[Submit](#)

Click on Submit or Save

Milestone 2: Creation of import set Tabel and Transform Map

Create Importset Table

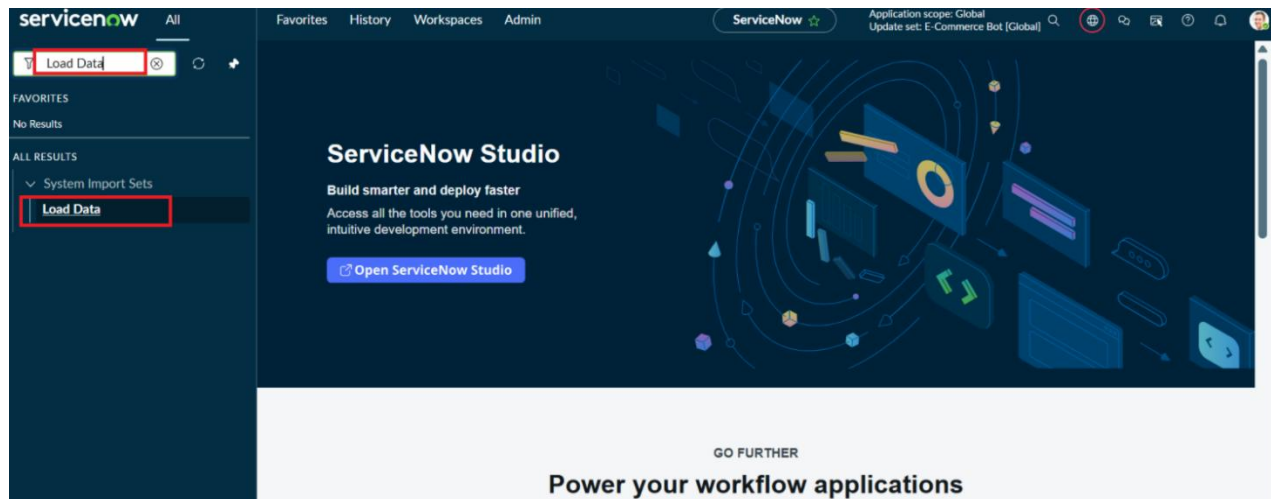
Import Set Table

The **Import Set Table** is a temporary staging table used to store employee data uploaded from the spreadsheet. It contains fields such as **Employee ID, Name, Email, Department, and Location**. The imported data is stored in this table before it is validated and transformed into the final employee records.

Navigation:

Go to Application Navigator

Load Data

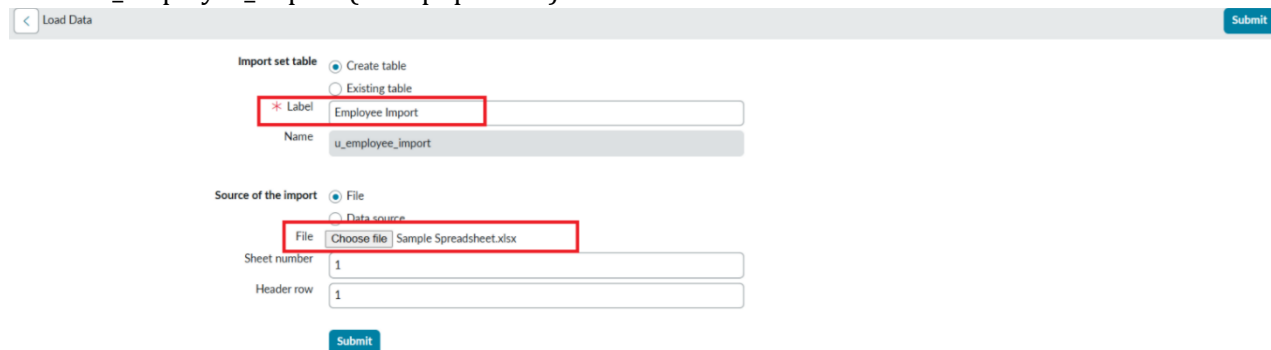


Click on Load Data and create an import table

Fill details as shown below

Label: Employee Import

Name: u_employee_import (auto populated)



Click on Submit

Once we click on Submit, we can see the success message as shown below.

Also we need to make sure to click on Create Transform Map.



Create Transform Map

Transform Map

The **Transform Map** defines how data from the Import Set Table should be transferred to the target Employee table. It maps fields such as **Employee ID, Name, Email, Department, and Location** from the imported data to the corresponding target fields. **Employee ID** can be used as the coalesce field to update an existing employee record or create a new record when the employee does not already exist.

Give inputs as shown below.

Name: Sample Spreadsheet Import

Source Table: Employee Import is auto-populated

Target Table: Employee Test - as we created earlier

Click on Auto Map Matching fields.

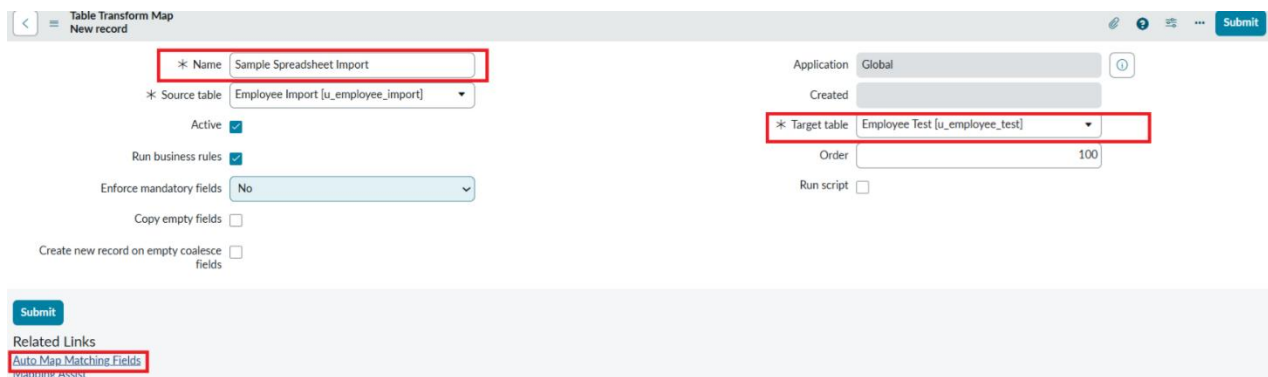


Table Transform Map
New record

* Name: Sample Spreadsheet Import

* Source table: Employee Import [u_employee_import]

Active: ☒

Run business rules: ☒

Enforce mandatory fields: No

Copy empty fields: ☐

Create new record on empty coalesce fields: ☐

Application: Global

Created:

* Target table: Employee Test [u_employee_test]

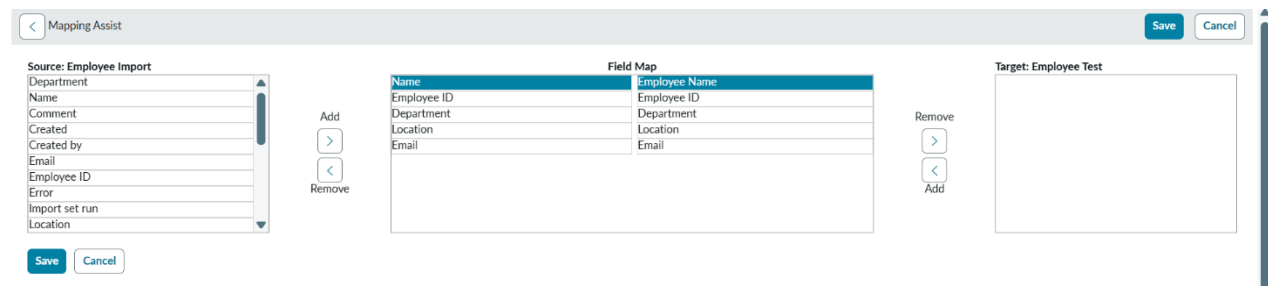
Order: 100

Run script: ☐

Submit

Related Links

[Auto Map Matching Fields](#)



Mapping Assist

Source: Employee Import

Field
Department
Name
Comment
Created
Created by
Email
Employee ID
Error
Import set run
Location

Add: > Remove: <

Field Map

Source	Target
Name	Employee Name
Employee ID	Employee ID
Department	Department
Location	Location
Email	Email

Remove: > Add: <

Target: Employee Test

Save **Cancel**

Click on Save

Make sure the both Source & Target table fields are mapped as shown below.

Table Transform Map
Sample Spreadsheet Import

Create new record on empty coalesce fields ☐

Copy Update Delete

Related Links
[Auto Map Matching Fields](#)
[Mapping Assist](#)
[Validate Coalesce Fields](#)
[Transform](#)
[Index Coalesce Fields](#)
[Run Point Scan](#)
[\[SN Utils\] Versions \(1\)](#)

Field Maps (5) Transform Scripts

Source field	Target field	Coalesce
u_name	u_employee_name	false
u_employee_id	u_employee_id	false
u_department	u_department	false
u_location	u_location	false
u_email	u_email	false

Now click on Transform as shown below

Table Transform Map
Sample Spreadsheet Import

Enforce mandatory fields Run script ☐

Copy empty fields ☐

Create new record on empty coalesce fields ☐

Copy Update Delete

Related Links
[Auto Map Matching Fields](#)
[Mapping Assist](#)
[Validate Coalesce Fields](#)
[Transform](#)
[Index Coalesce Fields](#)
[Run Point Scan](#)
[\[SN Utils\] Versions \(1\)](#)

Field Maps (5) Transform Scripts

Source field	Target field	Coalesce
u_name	u_employee_name	false
u_employee_id	u_employee_id	false
u_department	u_department	false
u_location	u_location	false
u_email	u_email	false

Now click on Transform link as shown below

Specify Import set and Transform map

* Import set

Available maps

Selected maps, run in order
Sample Spreadsheet Import - u_employee_test

Transform

After clicking Transform, make sure you see the success message as shown below.

Progress

Name	Transforming: ISET0010001
State	Complete
Completion code	Success
Message	Transformation complete

Next steps...

[ISET0010001](#) Go to the import sets for this data load

[Transform history](#) Show the transform history, related errors and log

[Import log](#) View the import log

Milestone 3: Transform Data & Validate and enable coalesce to avoid duplicate records

Transform Data & Validate

Description

Transform Data

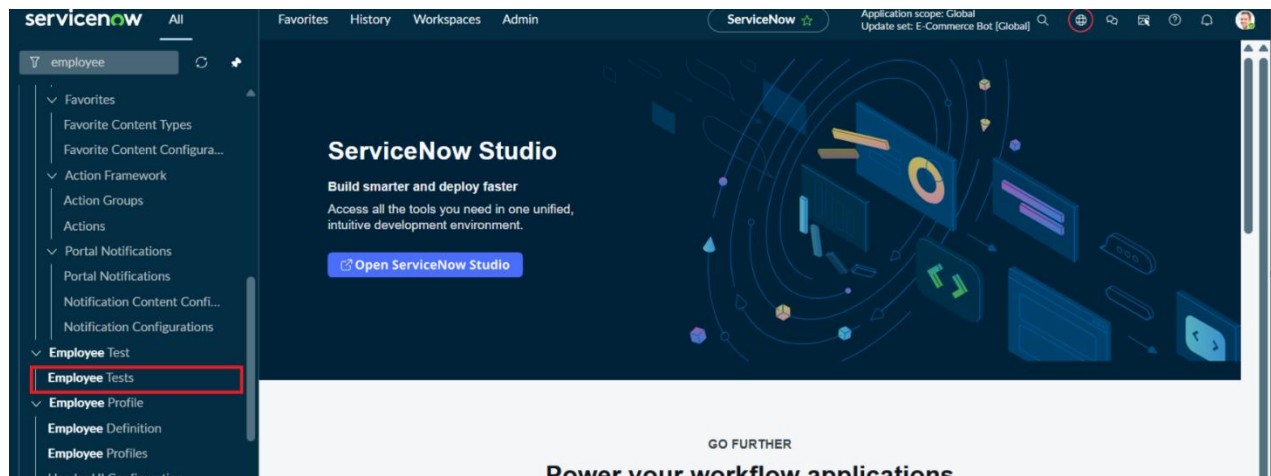
The **Transform Data** process transfers employee information from the Import Set Table to the target Employee table using the Transform Map. It maps and converts fields such as **Employee ID, Name, Email, Department, and Location** and ensures that the imported data is stored correctly in the target table.

Validate Data

The **Validate Data** process checks the imported employee information before it is added to the target table. It ensures that required fields such as **Employee ID, Name, and Email** contain valid data and helps identify incorrect or incomplete records so they can be corrected.

Navigation:

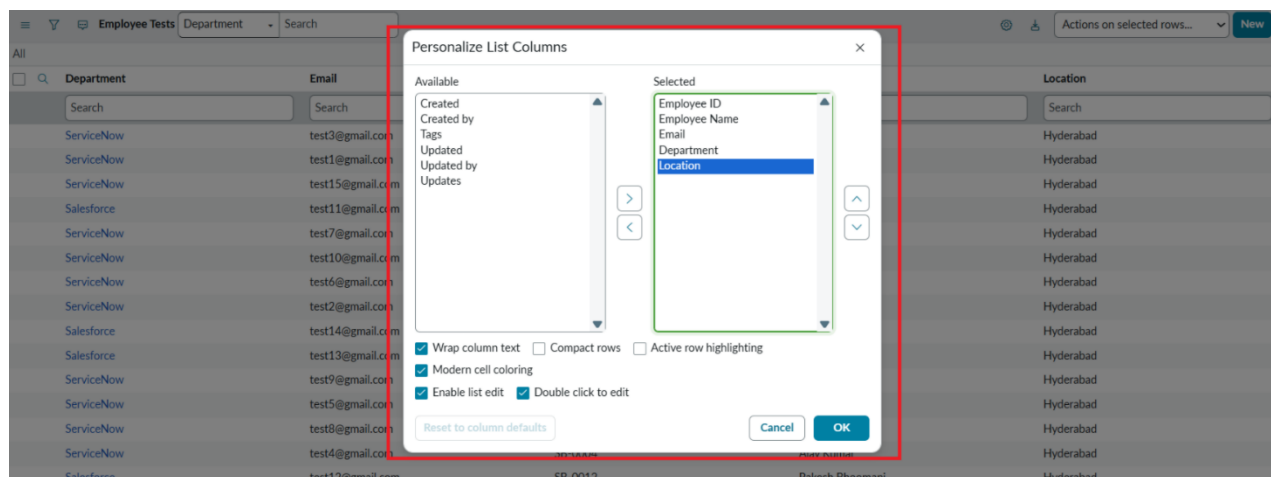
Open Employee Test Table from application navigator as shown below.



As our import got succeeded, all the records are created in the Employee Test Table.

Department	Email	Employee ID	Employee Name	Location
ServiceNow	test3@gmail.com	SB-0003	Meghana Katoju	Hyderabad
ServiceNow	test1@gmail.com	SB-0001	Ratan Kumar	Hyderabad
ServiceNow	test15@gmail.com	SB-0015	Srinu Nandipalli	Hyderabad
Salesforce	test11@gmail.com	SB-0011	Abhishek S	Hyderabad
ServiceNow	test7@gmail.com	SB-0007	Yaswanth Nikku	Hyderabad
ServiceNow	test10@gmail.com	SB-0010	Ravi Teja	Hyderabad
ServiceNow	test6@gmail.com	SB-0006	Sushma Ganji	Hyderabad
ServiceNow	test2@gmail.com	SB-0002	Sandeep Gujja	Hyderabad
Salesforce	test14@gmail.com	SB-0014	Sushanth J	Hyderabad
Salesforce	test13@gmail.com	SB-0013	Mounika K	Hyderabad
ServiceNow	test9@gmail.com	SB-0009	Praneeth S	Hyderabad
ServiceNow	test5@gmail.com	SB-0005	Bitla Sai Prasad	Hyderabad
ServiceNow	test8@gmail.com	SB-0008	Prashanth Reddy	Hyderabad
ServiceNow	test4@gmail.com	SB-0004	Ajay Kumar	Hyderabad
Salesforce	test12@gmail.com	SB-0012	Rakesh Bhoomani	Hyderabad

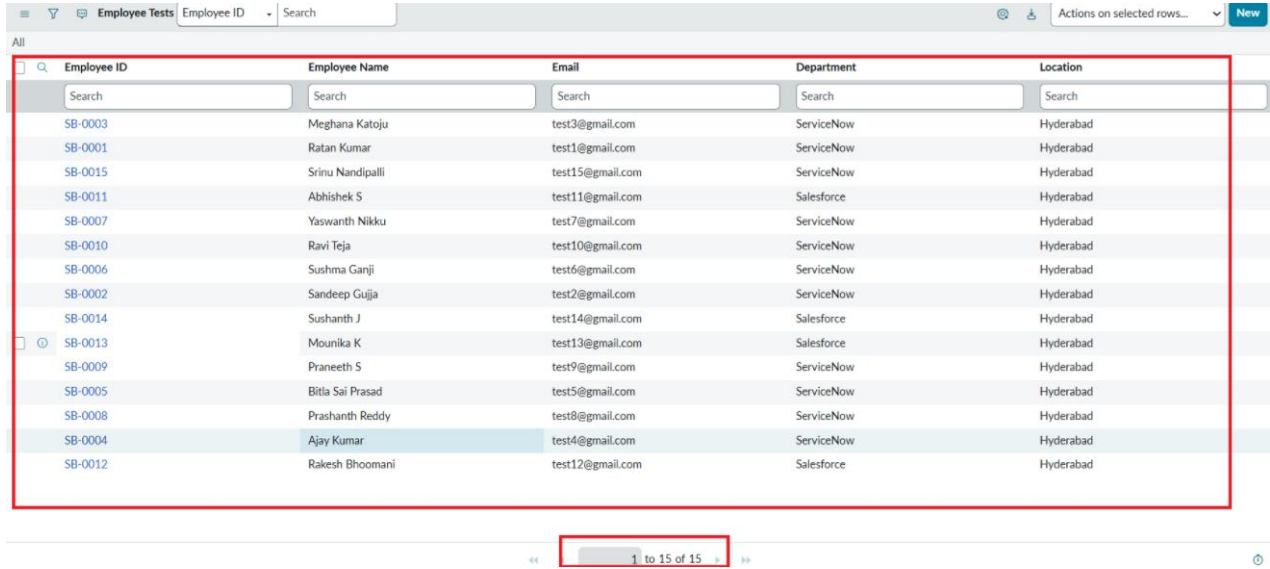
Make sure you rearrange the fields accordingly by using Personalised List Columns



Click on OK

Final Result:

All the records are successfully imported into the ServiceNow system with Import Set, Transform Map.



Employee ID	Employee Name	Email	Department	Location
SB-0003	Meghana Katoju	test3@gmail.com	ServiceNow	Hyderabad
SB-0001	Ratan Kumar	test1@gmail.com	ServiceNow	Hyderabad
SB-0015	Srinu Nandipalli	test15@gmail.com	ServiceNow	Hyderabad
SB-0011	Abhishek S	test11@gmail.com	Salesforce	Hyderabad
SB-0007	Yaswanth Nikku	test7@gmail.com	ServiceNow	Hyderabad
SB-0010	Ravi Teja	test10@gmail.com	ServiceNow	Hyderabad
SB-0006	Sushma Ganji	test6@gmail.com	ServiceNow	Hyderabad
SB-0002	Sandeep Guja	test2@gmail.com	ServiceNow	Hyderabad
SB-0014	Sushanth J	test14@gmail.com	Salesforce	Hyderabad
SB-0013	Mounika K	test13@gmail.com	Salesforce	Hyderabad
SB-0009	Praneeth S	test9@gmail.com	ServiceNow	Hyderabad
SB-0005	Bitla Sai Prasad	test5@gmail.com	ServiceNow	Hyderabad
SB-0008	Prashanth Reddy	test8@gmail.com	ServiceNow	Hyderabad
SB-0004	Ajay Kumar	test4@gmail.com	ServiceNow	Hyderabad
SB-0012	Rakesh Bhoomani	test12@gmail.com	Salesforce	Hyderabad

Enable Coalesce to Avoid Duplicate Records

Description

In real projects, the same employee data may be imported multiple times. To prevent duplicate employee records, we use **Coalesce**. The Coalesce will update the same records with new values if we add and the duplicate values are not inserted.

Steps:

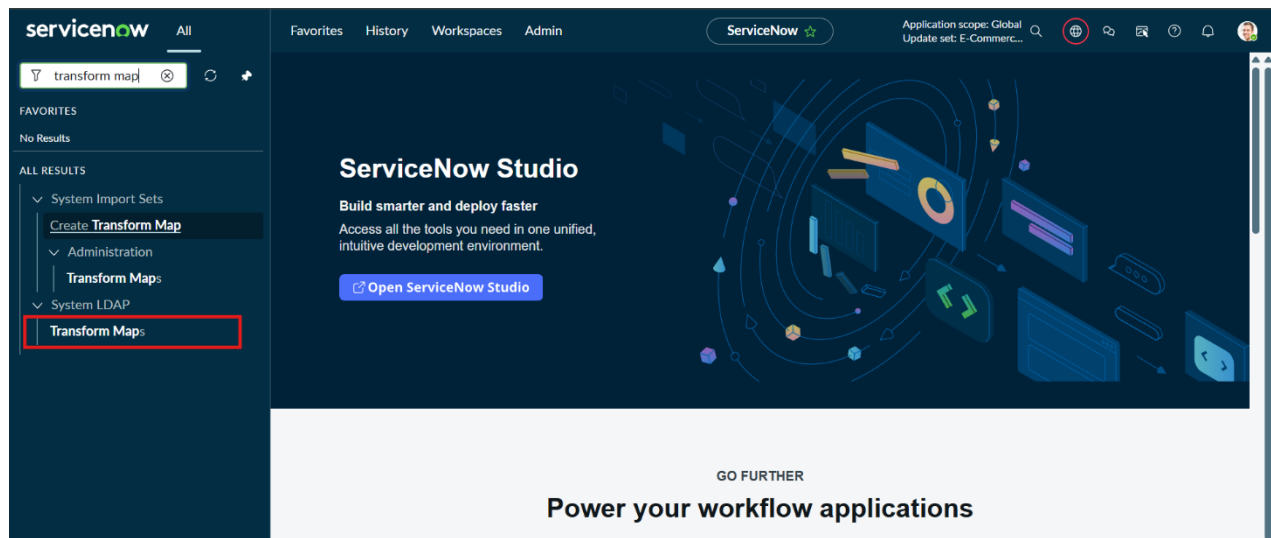
Open Transform map Under System import sets On All menus.

Open the Created Transform map , Name is Sample Spreadsheet Import.

Under Related list, select the Field maps Section and Make any one field Coalesce True which appears in False.

Now save the form.

Then again add some data which is in Excel format.



Open Transform map Which You are Created

Table Transform Maps Updated Search							
All							
☐	Name	Source table	Target table	Run business rules	Order	Active	Updated
☐	Sample Spreadsheet Import	Employee Import [u_employee_import]	Employee Test [u_employee_test]	true	100	true	2026-01-19 03:46:08
☐	Employee Job Profile Transform Map	Employee Job Profile Staging [sn_employee_position_stg]	Employee Job Profile [sn_employee_position]	true	100	true	2023-02-21 21:00:47
☐	Target Import	Target Import Set [imp_sn_gf_goal_target]	Target [sn_gf_goal_target]	true	100	true	2022-06-29 08:41:47
☐	Goal Import	Goal Import Set [imp_sn_gf_goal]	Goal [sn_gf_goal]	true	100	true	2022-06-29 06:06:40
☐	Google Example	label [oidc_google_import]	User [sys_user]	true	100	true	2018-02-06 13:42:44
☐	Okta Example	label [oidc_okta_import]	User [sys_user]	true	100	true	2018-02-02 14:21:44
☐	Azure AD Example	label [oidc_azure_ad_import]	User [sys_user]	true	100	true	2018-02-02 14:18:38
☐	Auth0 Example	label [oidc_auth0_import]	User [sys_user]	true	100	true	2018-02-02 14:04:26
☐	ADFS Example	label [oidc_adfs_import]	User [sys_user]	true	100	true	2018-02-02 10:58:24

Go to field mapping

Table Transform Map
Sample Spreadsheet Import

* Name: Sample Spreadsheet Import

* Source table: Employee Import [u_employee_import]

Active: ☒

Run business rules: ☒

Enforce mandatory fields: No

Copy empty fields: ☐

Create new record on empty coalesce fields: ☐

Application: Global

Created: 2026-01-19 03:46:08

* Target table: Employee Test [u_employee_test]

Order: 100

Run script: ☐

Copy Update Delete

Related Links

- [Auto Map Matching Fields](#)
- [Mapping Assist](#)
- [Validate Coalesce Fields](#)
- [Transform](#)
- [Index Coalesce Fields](#)
- [Run Point Scan](#)
- [\[SN Utils\] Versions \(1\)](#)

Field Maps (5) Transform Scripts Unused coalesce fields for empty values (1)

Source field	Target field	Coalesce

Go to coalesce field make true for one unique field

Related Links

- [Auto Map Matching Fields](#)
- [Mapping Assist](#)
- [Validate Coalesce Fields](#)
- [Transform](#)
- [Index Coalesce Fields](#)
- [Run Point Scan](#)
- [\[SN Utils\] Versions \(1\)](#)

Field Maps (5) Transform Scripts Unused coalesce fields for empty values (1)

Source field	Target field	Coalesce
u_name	u_employee_name	false
u_employee_id	u_employee_id	true
u_department	u_department	false
u_location	u_location	false
u_email	u_email	false

1 to 5 of 5

Inserting New Data In Excel Format

Description

Steps :

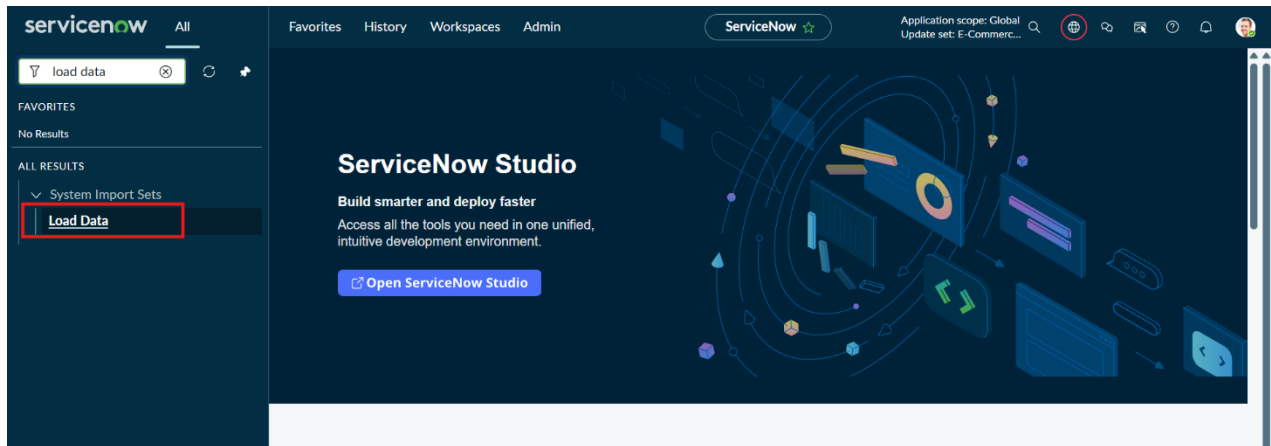
Navigate to All menus.

Search for Load data Under System Import set Open Load Data.

Fill the details as

Import set table as select Existing table {Employee Import}.

Now Click submit.



<

Load Data

Submit

Import set table

☐ Create table

☒ Existing table

Import set table

Employee Import [u_employee_import]

Source of the import

☒ File

☐ Data source

File

Choose File spreadsheetproject.xlsx

Sheet number

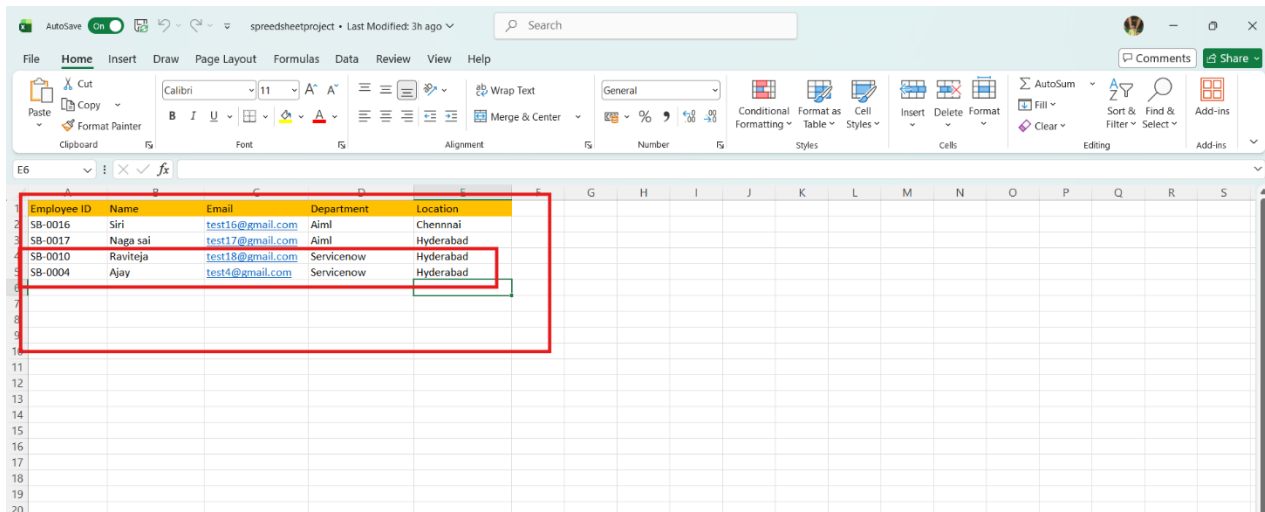
1

Header row

1

Submit

Newly Created Excel Data is this.



Employee ID	Name	Email	Department	Location
SB-0016	Siri	test16@gmail.com	Aiml	Chennai
SB-0017	Naga sai	test17@gmail.com	Aiml	Hyderabad
SB-0010	Raviteja	test18@gmail.com	ServiceNow	Hyderabad
SB-0004	Ajay	test4@gmail.com	ServiceNow	Hyderabad

Milestone 4: Creation Of reports & dashboards

Description

The Reports and Dashboards module provides a visual overview of the employee data imported into ServiceNow. Reports can be created to display employee information based on Department, Location, and other relevant fields, while the dashboard combines these reports into a single view. This helps administrators quickly monitor employee records, analyze data, identify trends, and make better decisions based on the imported information.

Description

Create Reports on Employee Test Table

Navigation Steps :

Open All menus, Search for **Reports**.

Under **Usage and governance** Select Reports.

Click the **New** button to create a new report.

Create **3 Reports** as follows.

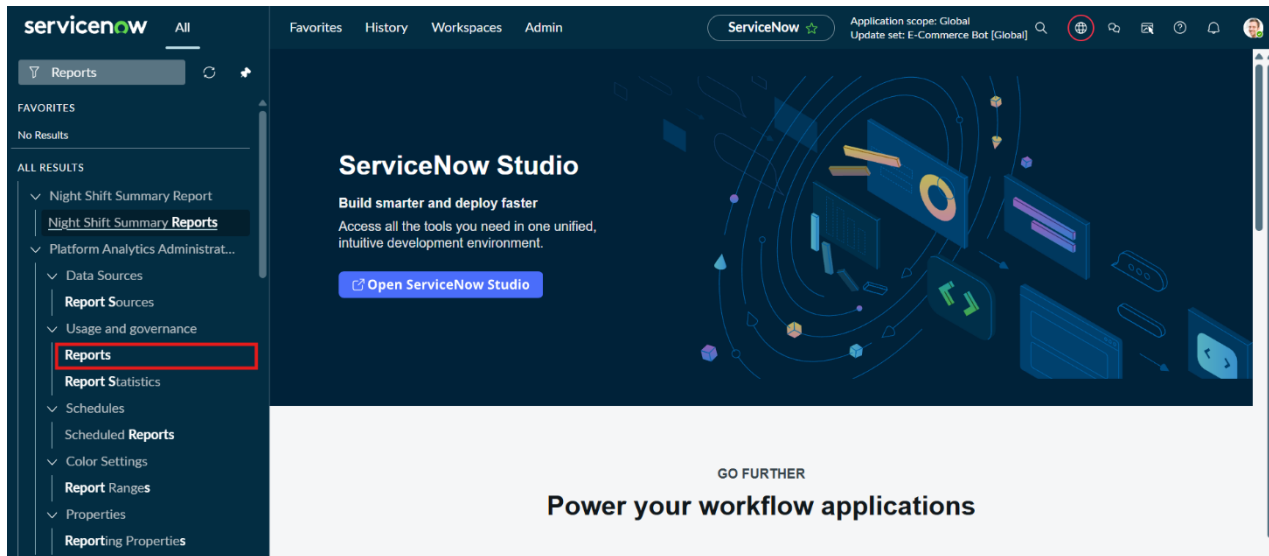
Report 1: Employees by Department

In Data

Report name : **Employees by Department.**

Source Type : **Table.**

Table : **Employee Test**, Now click the Next button.



Reports						
Title	Table	Type	Field Name	Created by	Updated	
SLA Performance Trend	Incident [incident]	Bar	opened_at	admin	2026-01-02 02:32:06	
Pending Justifications	Incident [incident]	List		admin	2026-01-02 01:13:00	
Top Assignees with SLA Violations	Incident [incident]	Bar	assigned_to	admin	2026-01-01 22:54:58	
Acknowledged vs Unacknowledged Breaches	Incident [incident]	Pie	u_sla_acknowledged	admin	2026-01-01 22:47:18	
SLA Breaches by Assignment Group	Incident [incident]	Bar	assignment_group	admin	2026-01-01 22:23:49	
SLA Breaches by Reason	Incident [incident]	Pie	u_sla_breach_reason	admin	2026-01-01 22:15:34	
Compliant CI Trend	CMDB Health Scorecard [cldb_health_scorecard]	Line		admin	2025-12-17 05:52:39	
Release Request for Asset Release of Service	Release Request [release_request]	List		admin	2025-12-17 05:52:39	

< Create a report Save Run

Data > Type > Configure > Style

* Report name
 Employees by Department

* Source type
 Table

* Table
 Employee Test [u_employee_test]

Description
 There is no description for this table. To add a

* Report Title : Employees by Department

Type a question about your data
 What do you want to see? Ask [How can I improve my results?](#)

To modify the current report, use the left panel or [Edit Condition](#).

Table: Employee Test [u_employee_test]

▼ All

Employees by Department

Now In Configure Select Group by Department Field.

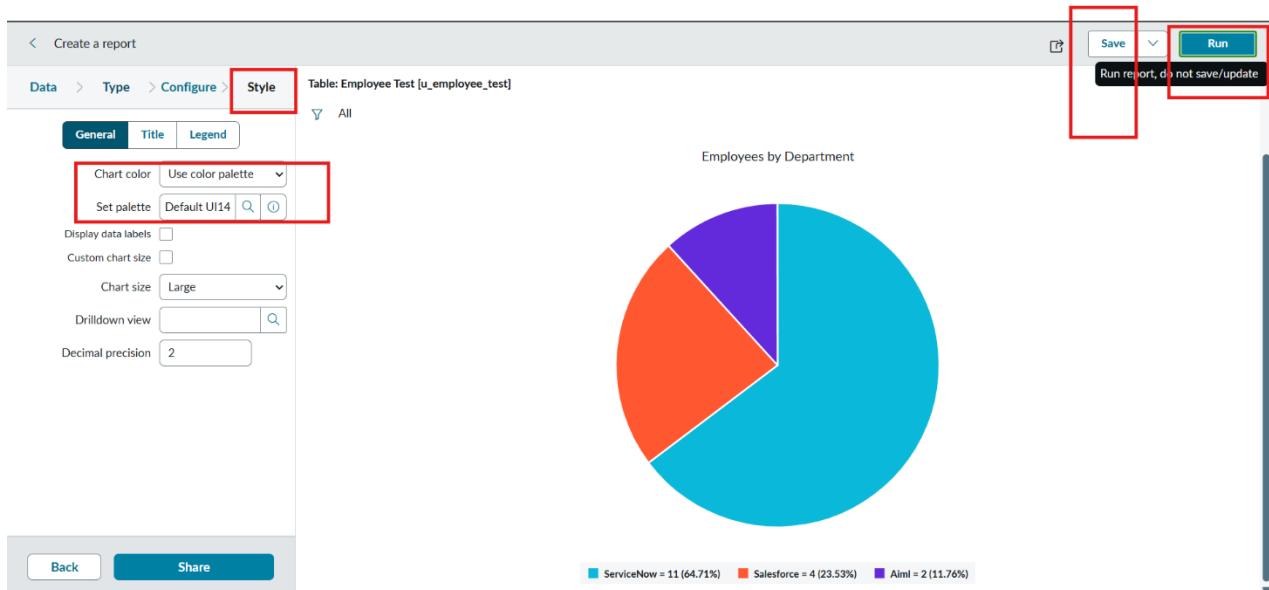
In the Aggregation Field Select Count.

Then Click next.

Now In Style Leave the Chart color by **default**.

And Click the **Run** button to Execute the given data Report.

And Click **save** button to Save the Report.



Adding Reports to Dashboard

Description

Create Dashboard

Navigation :

Open All menus → Search for Dashboards.

Under Self Service → Click Dashboards.

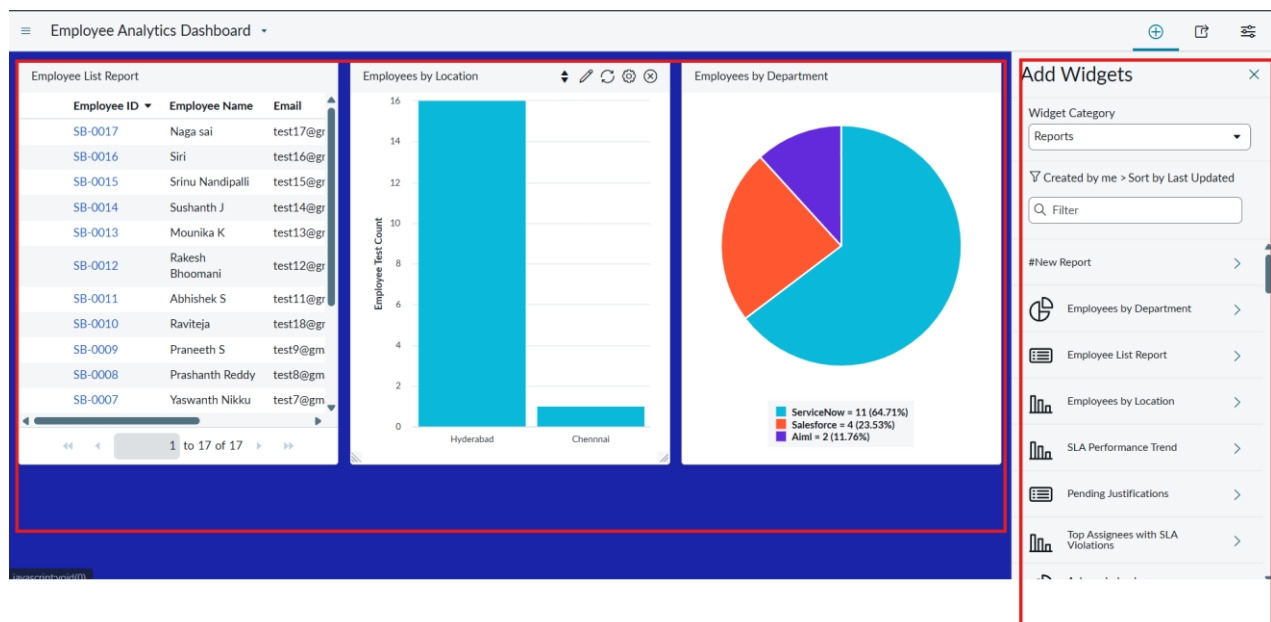
On the top right corner Click Create Dashboards or

You can create Directly by using the backend name[PA_DASHBOARDS.FORM].

Then Click New to Create new Dashboards.

Only the backend name will open the form if the ACL has the pa_dashboards access to Admin Overrides True only.

Then only we have access to the form



Conclusion:

The project successfully provides a structured and efficient approach for managing employee data using the **ServiceNow platform**. Employee information is imported from a spreadsheet using **Import Sets and Transform Maps** and stored in the **Employee Test table** for easy management and updating. The project also includes **Employees by Department, Employee by Location, and Employee List reports** to analyze and view employee information effectively. An **Employee Analytics Dashboard** brings these reports together and provides a clear visual representation of the data. By reducing manual data entry and organizing employee information in a centralized system, the project helps improve data accuracy, save time, and make employee records easier to access and monitor. Overall, the project demonstrates the effective use of ServiceNow features for employee data import, management, reporting, and dashboard-based analytics.